

Competing for molecules

Network Optimization of Cross-Industry Allocation

Master Thesis



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Date, year

By

Author

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This thesis has been prepared over six months at the Section for Indoor Climate, Department of Civil Engineering, at the Technical University of Denmark, DTU, in partial fulfilment for the degree Master of Science in Engineering, MSc Eng.

It is assumed that the reader has a basic knowledge in the areas of statistics.

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Abstract

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1 Introduction

The accelerating transition toward low-carbon energy systems has intensified research into the global production, transport, and allocation of green molecules such as green hydrogen, ammonia, e-methanol, and direct reduced iron (DRI). Existing modelling approaches span sector-specific optimisation, multi-carrier energy-trade modeling, and global commodity-flow assessments. However, these tools largely share a common characteristic: they assume cost-optimising behaviour, often embedded within political or economic conditions of individual countries, and therefore do not provide a neutral system-level view of how green molecules might distribute purely based on production availability and demand. The present study develops a model intended to fill this gap by exploring global flow patterns independent of national strategies, financing conditions, or trade preferences, focusing instead on the physical interaction between supply and demand under an open-market representation.

We are interested in understanding how green fuel is distributed between its production site and the various industries that purchase it for use. To illustrate this, we have chosen to focus on a small section of the world. This means that the conclusions presented in the report are not final, as not all production sites or all industries are represented. However, all conclusions in the report are made with the purpose of demonstrating that the model is correct and to show what it would be capable of with the correct input data.

1.1 Problem Statement

2 Theory

2.1 Linear Programming

Linear programming (LP) is a mathematical optimisation technique used to find the best outcome in a model whose requirements and objective are represented by linear relationships. A linear program consists of a linear objective function, which is either minimised or maximised, subject to a set of linear equality or inequality constraints, and a set of non-negativity conditions on the decision variables.

The general form of a linear programme is:

$$\min \quad \mathbf{c}^\top \mathbf{x} \quad (2.1)$$

$$\text{subject to} \quad A\mathbf{x} \leq \mathbf{b} \quad (2.2)$$

$$\mathbf{x} \geq \mathbf{0} \quad (2.3)$$

where $\mathbf{x} \in \mathbb{R}^n$ is the vector of decision variables, $\mathbf{c} \in \mathbb{R}^n$ is the cost vector, $A \in \mathbb{R}^{m \times n}$ is the constraint matrix, and $\mathbf{b} \in \mathbb{R}^m$ is the right-hand side vector. The feasible region defined by the constraints forms a convex polytope, and the optimal solution, if it exists, is attained at a vertex of this polytope [Integer bog].

Linear programming has a wide range of applications across logistics, supply chain management, energy systems, and resource allocation. A key property of linear programmes is that they are computationally tractable even at large scale, provided the problem structure is exploited effectively. When decision variables are further restricted to integer values, the problem becomes a mixed-integer linear program (MILP), which is generally harder to solve but captures a broader class of real-world decisions.

2.2 Network Models

Many optimisation problems can be naturally represented as networks, in which entities and their interactions are modelled as graphs. A graph $G = (N, A)$ consists of a set of nodes N and a set of arcs $A \subseteq N \times N$, where each arc (i, j) connects node i to node j . Arcs may be directed, meaning flow is permitted only from i to j , or undirected, meaning flow may travel in either direction.

Nodes typically represent locations, facilities, or decision points, while arcs represent connections such as roads, pipelines, or shipping routes. Each arc may carry associated parameters such as a cost c_{ij} , a capacity u_{ij} , or a length d_{ij} . Network models are a subclass of linear programmes with a particularly sparse and structured constraint matrix, which allows them to be solved very efficiently [Flow bog].

Well-known problems that can be formulated as network models include:

- **The shortest path problem:** finding the minimum-cost path between two nodes in a graph.

- **The minimum cost flow problem:** determining how to route flow through a network at minimum cost while respecting supply, demand, and capacity constraints.
- **The maximum flow problem:** finding the maximum feasible flow from a source node to a sink node subject to arc capacities.
- **The transportation problem:** allocating supply from a set of origins to a set of destinations at minimum cost.

These problems share a common structure and are all instances of the more general minimum cost flow problem, which is discussed further in the following section.

2.3 Shortest Path and Flow Problems

2.3.1 The Shortest Path Problem

The shortest path problem asks for the minimum-cost directed path from a source node s to a destination node t in a graph $G = (N, A)$ with arc costs c_{ij} for each arc $(i, j) \in A$. It can be formulated as a linear programme by introducing a flow variable $x_{ij} \in \{0, 1\}$ for each arc, indicating whether the arc is included in the path:

$$\min \sum_{(i,j) \in A} c_{ij} x_{ij} \quad (2.4)$$

$$\text{subject to } \sum_{j:(i,j) \in A} x_{ij} - \sum_{j:(j,i) \in A} x_{ji} = \begin{cases} 1 & \text{if } i = s \\ -1 & \text{if } i = t \\ 0 & \text{otherwise} \end{cases} \quad \forall i \in N \quad (2.5)$$

$$x_{ij} \geq 0 \quad \forall (i, j) \in A \quad (2.6)$$

The constraint in Equation 2.5 is the flow conservation constraint, ensuring that one unit of flow originates at s , is conserved at all intermediate nodes, and terminates at t . Due to the totally unimodular structure of the constraint matrix, the LP relaxation always yields integer solutions, meaning the integrality constraint on x_{ij} is automatically satisfied [flow bog].

2.3.2 The Transportation Problem

The transportation problem is a classical network flow problem that asks how to distribute a commodity from a set of supply nodes (origins) to a set of demand nodes (destinations) at minimum total cost. Each origin $i \in O$ has a fixed supply s_i and each destination $j \in D$ has a fixed demand d_j . Letting x_{ij} denote the amount shipped from origin i to destination j at unit cost c_{ij} , the problem is formulated as:

$$\min \sum_{i \in O} \sum_{j \in D} c_{ij} x_{ij} \quad (2.7)$$

$$\text{subject to } \sum_{j \in D} x_{ij} = s_i \quad \forall i \in O \quad (2.8)$$

$$\sum_{i \in O} x_{ij} = d_j \quad \forall j \in D \quad (2.9)$$

$$x_{ij} \geq 0 \quad \forall i \in O, j \in D \quad (2.10)$$

The transportation problem is a special case of the minimum cost flow problem in which there are no transshipment nodes, but only pure supply and pure demand nodes. While this structure is simple, it captures the essential trade-off between supply allocation and transport cost that underlies many logistics problems [Flow bog].

The green ammonia allocation modelled in this thesis shares the same fundamental structure: production sites act as supply nodes and offtake centres act as demand nodes. However, the presence of intermediate transit nodes, multiple transport modes, and capacity constraints means the problem is more general than a transportation problem, and is more accurately characterised as a minimum cost flow problem, described in the following subsection.

2.3.3 The Minimum Cost Flow Problem

The minimum cost flow (MCF) problem generalises both the shortest path and transportation problems by allowing flow to be sent from multiple sources to multiple destinations through a capacitated network with intermediate transshipment nodes. Each node $i \in N$ has a supply or demand value b_i , where $b_i > 0$ denotes a supply node, $b_i < 0$ denotes a demand node, and $b_i = 0$ denotes a transshipment node. The formulation is:

$$\min \sum_{(i,j) \in A} c_{ij} x_{ij} \quad (2.11)$$

$$\text{subject to} \quad \sum_{j:(i,j) \in A} x_{ij} - \sum_{j:(j,i) \in A} x_{ji} = b_i \quad \forall i \in N \quad (2.12)$$

$$0 \leq x_{ij} \leq u_{ij} \quad \forall (i,j) \in A \quad (2.13)$$

where u_{ij} is the capacity of arc (i,j) . The MCF problem is of particular relevance to this thesis, as the green ammonia allocation can be interpreted as a network in which production sites supply ammonia, offtake centres demand it, and transit nodes serve as transshipment points. The objective of minimising total cost across the network is directly analogous to the MCF formulation above.

2.4 Solving Network and Linear Programming Problems

(ER I TVIVL OM DETTE AFSNIT ER LIDT OVERFØLDIGT- ISÆR BRANCH AND BOUND DELEN DA VI IKKE RIGTIG BRUGER DET SENERE OG AT VORES MODEL IKKE ER EN HELTALSMODEL??)

2.4.1 Algorithms and Heuristics

Optimisation problems can be approached through two broad classes of solution methods: exact algorithms and heuristics. Exact algorithms guarantee finding the globally optimal solution, but may require substantial computational effort for large or complex instances. Heuristics, by contrast, sacrifice the guarantee of optimality in favour of computational speed, producing good but not necessarily optimal solutions within practical time limits [Integer bog].

For linear programmes, the most widely used exact algorithm is the *simplex method*, which traverses the vertices of the feasible polytope along improving edges until no further improvement is possible. While the simplex method has exponential worst-case complexity,

it performs very efficiently in practice and remains the dominant method for LP solving. *Interior point methods* offer an alternative with polynomial worst-case complexity and are often preferred for very large-scale LP instances [LP_textbook].

For network flow problems specifically, the *network simplex algorithm* exploits the graph structure of the constraint matrix to solve MCF problems significantly faster than a general LP solver, making it the method of choice for large network instances [network_flows_textbook].

2.4.2 Branch and Bound

When decision variables are required to take integer values, the problem becomes a mixed-integer linear programme (MILP), and LP-based algorithms are no longer directly applicable. The *branch and bound* algorithm is the most widely used exact method for solving MILPs [Ref til integer bog].

The algorithm proceeds as follows. First, the LP relaxation of the MILP is solved by dropping the integrality constraints. If the solution is integer-valued, it is optimal. If not, a fractional variable $x_j = f$ is selected and the problem is *branched* into two subproblems by adding the constraints $x_j \leq \lfloor f \rfloor$ and $x_j \geq \lceil f \rceil$ respectively, partitioning the feasible region. Each subproblem is solved as an LP, and its objective value provides a *bound* on the best integer solution obtainable within that branch. Branches whose bounds are worse than the best known integer solution are *pruned* and discarded. The algorithm continues recursively until all branches have either been pruned or yielded integer solutions.

To illustrate, consider a simple example with two integer variables $x_1, x_2 \geq 0$:

$$\max \quad x_1 + x_2 \quad (2.14)$$

$$\text{subject to} \quad 2x_1 + x_2 \leq 4, \quad x_1 + 2x_2 \leq 4, \quad x_1, x_2 \in \mathbb{Z}^+ \quad (2.15)$$

The LP relaxation yields $x_1 = x_2 = \frac{4}{3}$, with objective value $\frac{8}{3} \approx 2.67$. Since this is fractional, the algorithm branches on, say, x_1 : one subproblem adds $x_1 \leq 1$ and the other adds $x_1 \geq 2$. Solving each LP and continuing to branch as needed eventually identifies the optimal integer solution $x_1 = x_2 = 1$ with objective value 2, at which point all remaining branches can be pruned.

The efficiency of branch and bound depends heavily on the quality of the bounds and the branching strategy employed. Modern solvers enhance the basic algorithm with techniques such as cutting planes, which tighten the LP relaxation, and primal heuristics, which provide good integer solutions early to accelerate pruning.

2.5 Gurobi

Gurobi is a state-of-the-art commercial mathematical optimisation solver capable of solving LP, MILP, and quadratic programming problems at large scale [Gurobi hjemmeside]. It is widely used in both academia and industry and is the solver that is used in this thesis.

Internally, Gurobi solves MILP problems using a branch and bound framework enhanced with a range of advanced techniques, including cutting plane generation, presolve routines to reduce problem size, and sophisticated primal heuristics to find high-quality integer solutions early in the search. For the LP relaxations solved at each node of the branch and bound tree, Gurobi employs both the simplex method and barrier (interior point) methods, selecting automatically based on problem characteristics [Gurobi hjemmeside].

Gurobi is therefore well-suited to solving the model developed in this thesis. While the model is not necessarily large or complex in scale, Gurobi was selected partly due to

available license access and partly because the model may grow in complexity in future extensions, making it prudent to adopt a capable solver from the outset.

3 Literature review

3.1 Green Ammonia

3.1.1 Current landscape

Today, ammonia production is overwhelmingly fossil-based. It is produced by reacting nitrogen from the air with hydrogen derived from fossil fuels, typically natural gas or coal, through the Haber-Bosch process. Current global annual consumption is around 190 million tonnes, with the fertiliser industry accounting for approximately 70% of this demand. Ammonia synthesis is responsible for around 0.5 Gt of CO_2 annually, corresponding to roughly 1% of the greenhouse gas emissions. [1].

Beyond fertilisers, ammonia has gained attention as a potential decarbonisation vector. Its high hydrogen content by weight and existing global trade infrastructure make it an attractive hydrogen carrier, enabling hydrogen produced in renewable-rich regions to be transported and reconverted elsewhere. This has brought ammonia into focus for new end-use sectors, for example maritime shipping and steel production, where direct electrification is technically difficult or economically unfeasible at scale. In addition, ammonia is being explored as a low-carbon fuel for power generation and long-distance transport, further expanding its potential role in the future energy system. [2]

As decarbonisation pressure intensifies, the share of ammonia produced from renewable electricity, so-called green ammonia, is expected to grow substantially. Demand is projected to nearly double by 2035, driven by both continued fertiliser demand and emerging energy applications such as maritime fuel and hydrogen transport [3].

In longer-term scenarios, growth could be even more pronounced: IRENA projects that global ammonia demand could reach around 688 million tonnes by 2050 in a 1.5°C-aligned pathway, more than three times current levels.

This highlights the importance of examining pathways for green ammonia in the context of increasing demand and uncertainty surrounding its future production.

3.1.2 Techno-economic considerations

As of today, green ammonia is not yet deployed at scale and remains an emerging technology, with significant implications for cost estimates. The price of fossil-based ammonia currently ranges between 110 and 340 USD per tonne, while production with carbon capture and storage increases costs to approximately 170 to 465 USD per tonne. Green ammonia production costs are significantly higher at 700 to 1400 USD per tonne depending on location and technology, but are expected to decline substantially, potentially reaching 300 - 500 USD per tonne by 2030 - 2050 in regions with favourable renewable resources [2].

This cost structure is dominated by the upstream hydrogen production step, which can account for up to 90% of total production costs [2]. The three principal cost drivers are therefore the price of renewable electricity, electrolyser capital costs, and system utilisation rates. In regions combining low-cost renewables with high capacity factors, green ammonia is expected to become increasingly cost-competitive over time as electrolyser technology matures and production scales up. However, the pace and geography of this cost decline are themselves highly uncertain, and techno-economic projections are sensitive to underlying assumptions about technology development and energy system conditions. The wide cost ranges reported in the literature reflect this uncertainty.

3.1.3 Demand

The demand for green ammonia is still uncertain and can be used in different applications. While some industries already rely on grey ammonia, such as the fertiliser industry, other industries could adopt green ammonia as a way of transitioning towards a more sustainable alternative.

Fertiliser industry

The fertiliser industry accounts for approximately 70% of global ammonia demand [1] and represents the most natural starting point for green ammonia adoption. Since ammonia is a direct and non-substitutable input to nitrogen-based fertilisers, demand in this sector is structurally high and relatively price-inelastic.

Steel industry

The steel industry is one of the hardest sectors to decarbonise and accounts for a significant share of global industrial CO₂ emissions. One emerging pathway is the use of hydrogen, potentially delivered via ammonia, as a reducing agent to replace coking coal in iron ore reduction.

In this case, ammonia acts as a hydrogen carrier: it is shipped to the production site and cracked back into hydrogen for use in the direct reduction process. The extent to which the steel sector will rely on green ammonia depends heavily on policy targets and the pace of technological transition.

Maritime shipping

Ammonia is one of the leading candidate fuels for decarbonising deep-sea maritime shipping, alongside methanol and hydrogen. Its existing global trade infrastructure and high energy density make it logistically attractive, and several major shipping companies are actively developing ammonia-fuelled vessels [3].

Unlike the fertiliser sector, shipping demand for ammonia is largely a function of regulatory drivers such as carbon pricing and emissions targets rather than a direct process requirement. As of now the absence of strong CO₂ pricing in shipping limits the near-term incentive to pay a premium for green fuels, though emerging regulatory frameworks could change this.

3.2 Challenges and Uncertainties

Despite the promising cost trajectory outlined above, several structural barriers must be overcome before large-scale green ammonia deployment becomes feasible.

On the supply side, realising projected cost reductions depends on a substantial build-out of electrolyser manufacturing capacity that does not yet exist. Supply chain bottlenecks, could delay the learning curve that underpins optimistic cost forecasts. Suitable production sites are also geographically concentrated in regions with abundant wind or solar resources, many of which are remote from existing industrial demand centres, adding logistical complexity to supply chain development.

On the demand side, green ammonia faces a classic coordination problem. Without sufficient and stable demand, producers cannot justify the capital expenditure required to build large-scale facilities. Without large-scale production, prices remain too high to attract widespread industrial adoption. Early off-take agreements and policy support are therefore critical to breaking this deadlock [3]. This is compounded by the fact that many potential end-users, particularly in fertiliser production, currently rely on cheap grey ammonia, and face limited near-term incentive to pay a green premium without regulatory pressure or contractual certainty.

Infrastructure presents a further challenge. While ammonia benefits from existing global shipping and port infrastructure, significant investment is still required in dedicated storage, bunkering facilities, and onshore distribution networks [4]. The toxicity of ammonia also imposes safety and regulatory constraints that add complexity and cost to its handling and transport, particularly in densely populated regions.

Finally, the policy environment remains deeply uncertain, as illustrated by recent developments in maritime shipping regulation. While the IMO adopted a revised GHG strategy in 2023 and approved a Net-Zero Framework in April 2025, formal adoption of the framework was postponed for one year in October 2025 following opposition from the United States, Saudi Arabia, and several major flag states [5]. The earliest possible entry into force is now 2029, introducing renewed uncertainty for fuel producers and investors. In the interim, the EU's FuelEU Maritime regulation remains the most substantive binding framework, but applies only within EU waters [6]. This regulatory fragmentation is itself a barrier to the green ammonia transition, as long-term investment in alternative fuel infrastructure requires policy certainty that is currently absent at the global level.

3.3 Previous work on the subject

Several optimisation models have been developed to analyse the supply and distribution of green ammonia and related hydrogen derivatives, though each differs in scope, methodology, and the questions it is designed to answer.

In the paper, *Optimal fuel supply of green ammonia to decarbonise global shipping* they present a cost minimisation model for supplying green ammonia to meet a proposed global shipping fuel demand in 2050. The model optimises infrastructure location, transport routes, and investment requirements across a global network of production and demand nodes, with the levelised cost of ammonia (LCOA) serving as the key metric for evaluating and comparing supply pathways.[7]

A strength in this model is the treatment of shipping-specific infrastructure and gives a clear indication of the large production facilities that are required for the large demand in the maritime sector to be fulfilled.

However, the model is specific to the maritime sector and does not consider competition for green ammonia across multiple industries.

The IRENA global hydrogen trade modelling effort [8] takes a similar perspective, exploring how hydrogen and hydrogen-derived commodities, could be produced and traded internationally by 2050. Its focus is on regional production potential, transport infrastructure costs, and the economics of different trade corridors. Like the model in, [7] it is primarily a planning tool and does not address how scarce supply would be allocated across competing demand sectors.

Our model differs from these in several important aspects. Rather than a optimal infrastructure planning model, it models the allocation of green molecules to different industries simultaneously. This allows us to examine how different demand nodes compete for available supply, which production sites are utilised, and where unmet demand concentrates geographically. The two approaches are complementary, infrastructure solutions derived from models such as [7] could serve as direct input to our framework, allowing us to evaluate how optimised supply chains perform when subjected to multi-industry competition across different scenarios.

Furthermore the model is designed as a flexible analytical tool where different future cases can be used as input. Due to the structure of the model industries or green molecules

can be introduced without restructuring the underlying framework, and policies, prices, and demand levels can be adjusted to reflect different assumptions about adoption rates, production capacity, or regulatory environments. This makes the model well-suited to examine a wide range of conditions, from scenarios of supply scarcity, where competition between industries determines who receives green ammonia and where unmet demand concentrates, to scenarios of over production, where the question shifts to which regions and production sites are most utilised and which are competed out by cheaper alternatives.

4 Modeling

The problem is conceptualised as a network model in which the system is represented by nodes connected through edges that carry flow.

4.1

P : set of production sites

T : set of transit points

O : set of off take centers

N : set of all nodes

To formulate the problem we start by defining a set for each type of node in the network. The network consists of production sites, transit points and finally off take centers, all connected through edges that define the transportation routes between the nodes. Let P denote the set of production sites, T denote the set of transition nodes, O the set of off take centers and N the full set of nodes in the network. All possible edges between all nodes are made and a cost matrix will define the cost for all edges (i, j) from node i to node j .

4.2

Each edge (i, j) is associated with a cost that refers to the cost of the transportation of green ammonia on the specific edge. We assume that the cost for each industry is the same hence no industry will pay more or less for the green ammonia that is being produced in each off take center. This makes the cost matrix only dependent on the nodes i and j . Edges between two transit nodes $t \in T$ are shipping routes, where edges between production sites $p \in P$ and transit nodes or from transit node to off take centers nodes $o \in O$ is routes onshore. The transportation cost is computed based on attributes such as the shipping rate, the geographical region of each node, the transportation mode used on that edge, the physical distance between nodes, and an estimate of the operating cost of onshore transportation. If transportation between two nodes are not possible the transportation cost will be set to ∞ . The cost matrix $c_{i,j}$ will therefore be a $N \times N$ matrix.

For each production site $p \in P$, the parameter $maxP^p$ specifies the maximum production capacity available. Similarly, each off take center $o \in O$ has a known demand level D^o , which the network attempts to satisfy through delivered quantities or, if necessary, unmet demand penalized through a parameter π . Since the problem is a minimization problem, we set π to be a high cost.

This will ensure that the model will distribute the green ammonia produced at the production sites before meeting the rest of the demand with the high cost. This is mostly effective in a scenario where there is less green ammonia produced than demand which is how it is today. Later when we introduce different scenarios this parameter π will be changed.

Each production site $p \in P$ has an associated production cost given by $Prodcost^p$, reflecting the cost of manufacturing one unit of green ammonia.

Together, these parameters describe the economic structure of the network and provide the data necessary to evaluate transportation, production, and shortage decisions within the model.

4.3

Furthermore we introduce several decision variables that describe flow, production level and demand fulfilment. For each production site $p \in P$ and for every edge (i, j) the variable $f_{i,j}^p$ denotes the quantity of ammonia produced at production site p shipped along the edge going from node i to node j . This variable ensures that it is possible to track where the green ammonia delivered to off take center $o \in O$ is produced. The variable $prod^p$ represents the total production of ammonia on production site $p \in P$. At the off take centers, the variable q^o specifies the amount of ammonia delivered to off take center $o \in O$, while u^o is the unmet demand at the off take center. All variables are constrained to be nonnegative. Together, these decision variables capture the essential operational choices of the system, enabling the model to determine optimal production levels, routing decisions, and the extent to which demand is met across the network.

4.4

We end up with the following model

$$\min \sum_{p \in P} \sum_{i \in N} \sum_{j \in N} c_{i,j} f_{i,j}^p + \text{Prodcost}^p \text{prod}^p + \sum_{o \in O} \pi u^o \quad (4.1)$$

$$\text{s.t.} \quad \sum_{j \in N} f_{p,j}^p = \text{prod}^p \quad \forall p \in P \quad (4.2)$$

$$\text{prod}^p \leq \text{MaxP}^p \quad \forall p \in P \quad (4.3)$$

$$\sum_{p \in P} \sum_{i \in N} f_{i,o}^p = q^o \quad \forall o \in O \quad (4.4)$$

$$q^o + u^o \geq D^o \quad \forall o \in O \quad (4.5)$$

$$\sum_{j \in N} f_{t,j}^p = \sum_{i \in N} f_{i,t}^p \quad \forall p \in P, \forall t \in T \quad (4.6)$$

$$f_{i,j}^p = 0 \quad \forall p \in P, \forall i \in P \setminus \{p\} : i \neq p, \forall j \in N \quad (4.7)$$

$$f_{i,j}^p \geq 0, \quad q^o \geq 0, \quad u^o \geq 0, \quad \text{prod}^p \geq 0 \quad (4.8)$$

The objective function represents the total economic cost of operating the network and is minimised to identify an efficient allocation of production and transportation resources. The objective function captures all cost elements that contribute to the marginal cost of delivering one unit of ammonia to each off take center node $o \in O$. Transportation costs are included through the edge specific coefficients $c_{i,j}$, ensuring that the model accounts for the logistical expense of moving green ammonia from production sites through transfer points and onward to off take centers. Production costs at each production site are incorporated through Prodcost_p , reflecting the cost of producing green ammonia at the production sites. Finally, a penalty cost is assigned to unmet demand. This term is essential because the total available supply of green ammonia is insufficient to satisfy all demand in the system. By incorporating unmet demand in the objective, the model explicitly recognises the scarcity of green ammonia and internalises the economic consequences of that scarcity.

First, production capacity at each production site is limited in the constraint 4.3. Together with constraint 4.2, it ensures that the amount of green ammonia shipped from each production site $p \in P$ do not exceed the maximum amount of green ammonia that each production site can produce. Constraint 4.6 ensures flow conservation in each transit point in the model. This flow conservation is based on where the green ammonia that is being transported is produced. Hence the incoming flow of ammonia produced at production site $p \in P$ is equal to the outgoing flow of ammonia produced at the same production site. This constraint takes all transit points $t \in T$ into consideration. Two constraints specify how delivered quantities and unmet demand are accounted for at the off take centers. Constraint 4.4 defines the delivered quantity q_o as the total flow arriving at each off take center node from all upstream nodes. Constraint 4.5 enforces that the sum of delivered and unmet demand must match the total demand at each off take center $o \in O$, ensuring accurate allocation of the produced green ammonia. Constraint 4.7 force the flow of ammonia from production site $p \in P$ on edges originating from any other production site to be zero. This ensure that green ammonia produced at production site $p \in P$ is only shipped from this production site. Lastly, all decision variables are continuous and nonnegative set in constraint 4.8.

Together, these constraints ensure that the model reflects the physical realities of production, transportation, and demand fulfillment in the network.

5 Data processing

5.1

The network in the model is constructed as a directed graph consisting of nodes and edges, where nodes represent either geographic regions or port locations and edges represent feasible transportation links between them. Three types of nodes are defined, each corresponding to a different role: production, offtake, and transit. The geographic scope of the model is divided into a set of regions, each representing a distinct area with either green ammonia production potential, industrial demand, or both. These regions were defined based on two key datasets provided by the MMCZCS. The first contains estimated shipping costs for ammonia transport between regions, which directly defines the maritime cost structure of the network [MMCZCS]. The second is a BCG analysis that provides regional breakdowns of production activity across industries such as fertiliser and steel, from which industry-specific demand at the regional level is derived [BCG]. Together, these datasets establish the geographic resolution of the model and determine where production and demand capacity are located. The regions and their respective areas are shown in Figure 5.1.

A regional rather than country-level approach is beneficial for this model, as the underlying data is already reported at a regional granularity. Disaggregating it to country level would require detailed data processing of each individual country's industrial production, which is not feasible within the scope of this thesis. Furthermore, the model is proposed as an analytical tool rather than a logistical planning tool, and its intended conclusions concern how the overall pattern of global supply and demand would shift in response to factors such as changing green ammonia production costs or the introduction of CO₂ taxation. Representing each country individually across every industry would substantially increase computational complexity without adding meaningful analytical value.

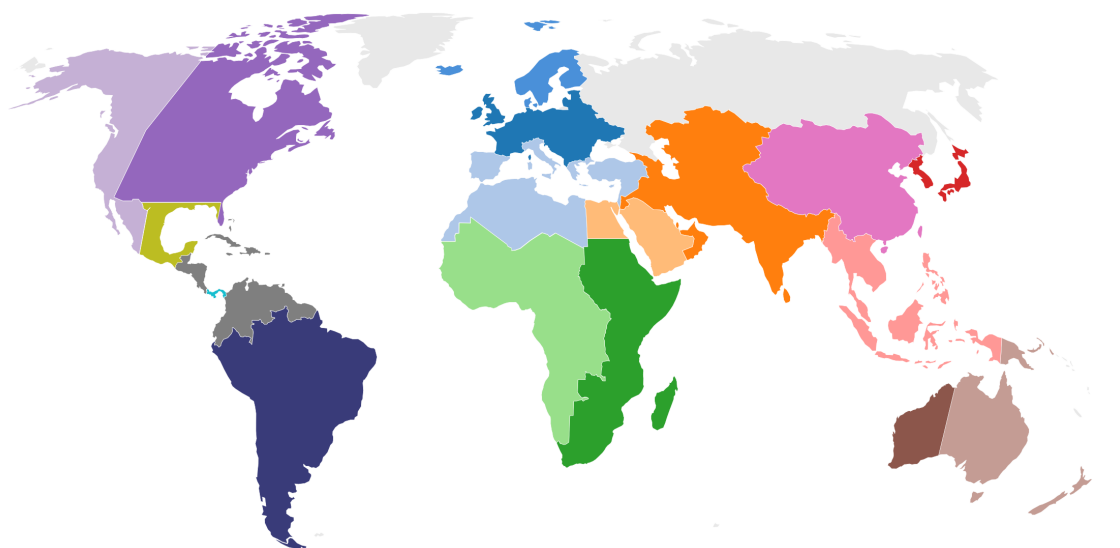


Figure 5.1: Region overview

Color	Region		
	Western Europe		East Africa
	Nordics		West Africa
	Mediterranean		NA East Coast
	Middle East & India		NA West Coast
	Red Sea		Gulf of Mexico
	Japan & South Korea		Panama
	China & Hong Kong		N Latin America
	South-east Asia		S Latin America
	West Australia		Other
	East Australia		

Table 5.1: Region color legend

5.2 Demand

The demand nodes represent regionally aggregated green ammonia demand sites, divided by industry, and serve as the offtake nodes for all flows in the network.

Fertiliser and Steel

Demand nodes for fertiliser and steel are derived from production capacity data published by BCG [BCG], which provides both global production volumes and their regional breakdown expressed as percentage shares. These shares correspond directly to the regional structure adopted in this model.

Within each region, the data specifies a set of representative city locations, which determine the number and placement of demand nodes assigned to that region. Although this model operates with aggregated regional demand, multiple nodes are placed within each region to reflect geographic diversity in industrial activity. Once total demand has been calculated at the regional level, it is distributed equally across all demand nodes within that region.

The methodology for translating sectoral production volumes into equivalent green ammonia demand varies by industry.

For the fertiliser sector, production volumes are converted directly into green ammonia demand. Since fertiliser production currently relies on grey ammonia, and green and grey ammonia are chemically identical, a direct one-to-one substitution is applied without any adjustment for energy content. The dominant nitrogen fertiliser produced globally is urea [9], which has an ammonia conversion factor of 0.58, meaning that 0.58 tonnes of ammonia are required to produce one tonne of urea.

For the steel sector, the ammonia demand intensity relative to total steel production is assumed to be 0.31, derived as follows. Hydrogen-based direct reduced iron (DRI) production requires approximately 0.055 tonnes of hydrogen per tonne of steel [10]. Since ammonia serves as the hydrogen carrier in this model, the required ammonia input is determined by the hydrogen mass fraction in ammonia [11]:

$$\frac{3}{17} \approx 0.176 \implies 1 \text{ tonne NH}_3 = 0.176 \text{ tonne H}_2$$

Dividing the hydrogen requirement by this mass fraction yields the ammonia intensity factor:

$$\frac{0.055}{0.176} \approx 0.31 \text{ tonnes NH}_3 \text{ per tonne of steel}$$

Once total green ammonia demand has been established for each sector at the global level, it is disaggregated to the regional level by applying the BCG regional production shares [BCG]. Within each region, demand is distributed evenly across the offtake centres located there, such that each centre receives an equal share of regional demand.

Shipping

Global marine bunkering volumes are projected to reach 249.65 Mt of fuel oil in 2026, reflecting a market that continues to grow year-on-year [global_top10]. Supply is highly concentrated: the ten largest bunkering ports collectively handled 111.9 Mt in 2024, representing approximately 45% of global volume, with Singapore alone accounting for 54.9 Mt, roughly 22% of the worldwide total.

Given this concentration, the ten largest bunkering ports are selected as the shipping demand nodes in this model [global_top10]. Total global shipping demand is then distributed across these ports in proportion to their respective bunkering volumes.

To assess the ammonia volumes required to replace conventional marine fuel oil on an energy-equivalent basis, the lower heating values (LHV) of the two fuels are compared. Marine fuel oil is a blend of various fuel fractions; a representative LHV of 41 MJ/kg is adopted here [Find ref]. Ammonia has an LHV of 18.6 MJ/kg [IEA roadmap]. The energy-equivalent substitution ratio is therefore:

$$\frac{18.6}{41} \approx 0.45 \implies 2.2 \text{ kg NH}_3 \text{ per kg of fuel oil}$$

meaning that approximately 2.2 kg of ammonia is required to deliver the same energy as 1 kg of marine fuel oil.

Unmet Demand and the Parameter π

Unmet ammonia demand is incorporated into the model to ensure feasibility in cases where projected green ammonia supply is insufficient to satisfy total global demand across all industries. A penalty cost π is assigned to each unit of unmet demand in the objective function, making unmet demand costly to the optimiser while still allowing the solver to find the best attainable solution under binding supply constraints.

The penalty parameter π is constructed by summing the highest observed production cost across all offtake centres and the highest applicable transportation cost, ensuring that π strictly exceeds any cost that could be incurred by supplying ammonia through the modelled network. This guarantees that the optimiser will always prefer to route supply through available production and transport rather than resort to unmet demand:

$$\pi = 1,500,000,000 + (2 \times 540,000,000 + 150,000,000) = 2,730,000,000 \quad (5.1)$$

where 1,500,000,000 represents the maximum production cost, and the transportation component reflects a journey passing through a transit node, incurring onshore transport costs twice, plus the maximum offshore shipping cost.

5.3 Production

Production nodes represent regionally aggregated green ammonia production sites and serve as the origin points for all flow in the network. An initial approach placed production nodes at specific coordinates sourced directly from the MMCZCS dataset[reference], with an estimate of production capacities. However, since all other nodes in the model are defined on a regional basis, and since green ammonia production at the scale considered here is entirely prospective rather than operational, a more consistent methodology was adopted to align production nodes with the regional aggregation approach used throughout the model.

Production locations are derived from the dataset published by [7], which estimates green ammonia production costs and capacities on a 1×1 degree global grid. The underlying model combines hourly weather reanalysis data for local wind and solar resources with cost parameters for renewable energy generation, electrolysis, and Haber-Bosch synthesis to compute a site-specific levelised cost of ammonia (LCOA). Production capacity at each site is constrained by land availability, assessed through satellite-based land cover classification with exclusions for protected areas and steep terrain.

As a preprocessing step, all grid cells with zero production capacity are removed. The remaining sites are aggregated into 50 production nodes using k -means clustering. K -means is a partitioning algorithm that assigns each data point to one of k clusters by minimising the total within-cluster squared distance between each point and its centroid, iterating between point reassignment and centroid updates until convergence [Ref til k-means]. The choice of 50 nodes ensures a number of production sites which are approximate the same number as demand nodes in the model, and also so the placement of production nodes can have the same approach as described in Section ???. Hence the production sites are not only one node per region, yielding a more realistic distribution of production capacity.

For each cluster, the production site with the lowest LCOA is selected as the representative node, and its coordinates are used to position the production node. The maximum production capacity of each node is set equal to the sum of capacities across all sites within the cluster.

Following clustering, nodes falling outside the model’s defined regions are removed. This includes several nodes located in Russia, which lies outside the geographic scope of the model, as well as two nodes in Greenland and one node on a remote Australian island, none of which represent realistic production locations within the modelled system.

Production costs at each node are based on regional estimates provided by the MMCZCS [ref til data], which reflect projected green ammonia production costs in 2030 and account for regional differences in renewable energy availability and infrastructure conditions. For nodes located in regions not covered by the MMCZCS dataset, the production cost is set equal to that of the nearest region for which data are available. The projected production costs across all modelled regions are summarised in Table 5.2. The share of global production assigned to each node is determined by expressing its maximum capacity as a fraction of total capacity across all

Table 5.2: Production cost by region (2030)

Region	Cost (\$/t)
Western Europe	1400
Nordics	1400
Mediterranean	1200
Middle East & India	1100
Red Sea	1350
Japan & South Korea	1100
China & Hong Kong	1100
South-east Asia	1100
West Australia	1100
East Australia	1050
East Africa	1500
West Africa	1500 ¹⁷
NA East Coast	1200
NA West Coast	1200
Gulf of Mexico	1150

production nodes. This percentage is then used to derive each node’s absolute production capacity when the total globally projected production volume is chosen.

Production costs at each node are based on regional estimates provided by the MMCZCS [ref til data], which reflect projected green ammonia production costs in 2030 and account for regional differences in renewable energy availability and infrastructure conditions. For nodes located in regions not covered by the MMCZCS dataset, the production cost is set equal to that of the nearest region for which data are available. The projected production costs across all modelled regions are summarised in Table 5.2.

5.4 Transportation

Transit nodes represent individual ports that serve as intermediary points in the maritime transportation network. They are selected from the NGA World Port Index [12] and filtered to include only ports classified as *Large*. Entries corresponding to offshore oil platforms are removed, as these cannot receive flow from production nodes via onshore transport and are therefore not suitable as transit nodes in this model.

Edges in the network define the feasible transportation links between nodes. All transit nodes are connected to all other transit nodes, representing maritime shipping routes between ports. Production nodes and off take centre nodes are connected to transit nodes and to each other within their region, representing onshore transport links.

The cost structure of all edges is captured in the transportation cost matrix C , which is an $N \times N$ matrix where each entry C_{ij} represents the cost of transporting one tonne of ammonia from node i to node j . The matrix is not symmetric, as transport is not necessarily feasible in both directions between a given pair of nodes. In particular, production nodes are source-only: ammonia can be transported *out* of a production node but never *into* one, meaning a production node cannot appear as destination j .

All node pairs for which transport is infeasible are assigned $C_{ij} = \infty$, ensuring that the optimiser will never route flow along such an edge. By default, edges are initialised between all $N \times N$ node pairs, and infeasibility of transportation on edges are then calculated according to the transport mode rules described in the following sections.

Offshore transport

Edges are defined between all transit nodes in the model, meaning that maritime transport between any two ports is considered feasible by default. The offshore cost in matrix C_{ij} is constructed from shipping cost data published by BCG [BCG], which provides the cost of transporting one tonne of ammonia between world regions by ship. Where a direct port-to-port cost is not available, the cost is approximated using the regional cost corresponding to the respective origin and destination regions. These region are the same as presented in Section 5.1.

It should be noted that no assumption is made regarding the fuel or vessel type used for these shipments. The shipping costs are adopted as given and do not reflect a transition to green ammonia-powered vessels.

Onshore transport

Several approaches were considered for calculating onshore distances between nodes in the model. Initially, OpenRouteService was used to obtain realistic truck routing distances between transit nodes, production sites, and offtake centres. However, this approach introduced two key limitations. First, routing results were not always available for all node pairs, as some coordinate combinations fell outside the coverage of the routable road network. Second, and more fundamentally, road-based routing is inherently specific to truck transport and cannot be meaningfully extended to other modes such as pipeline or rail, which do not follow road networks. To overcome these limitations, a more generalised distance metric was adopted, leading to the selection of the Haversine formula as a unified basis for all onshore distance calculations in the model.

The Haversine formula is a standard method for computing great-circle distances between two points on the Earth’s surface based solely on their latitude and longitude coordinates. The distance is determined by computing the central angle between the two points on a sphere and converting this angle into a linear distance. This makes the approach well-suited to applications where mode-specific network route data are either unavailable or not directly comparable across transport types. A similar distance-estimation approach has been applied in other studies modelling the transport of green ammonia and comparable energy carriers [insert source], demonstrating that great-circle distance calculations provide a robust and consistent basis for cost modelling across different transport modes.

The Haversine formula is given by:

$$d = 2r \arcsin \left(\sqrt{\sin^2 \left(\frac{\Delta\varphi}{2} \right) + \cos(\varphi_1) \cos(\varphi_2) \sin^2 \left(\frac{\Delta\lambda}{2} \right)} \right) \quad (5.2)$$

where d is the great-circle distance between the two points, r is the mean radius of the Earth, φ_1 and φ_2 are the latitudes of the two points, $\Delta\varphi$ is the difference in latitude, and $\Delta\lambda$ is the difference in longitude [Ref til Haversine function]. Within this model, Equation 5.2 is applied uniformly to generate consistent straight-line distance estimates for all onshore transport modes, ensuring that cost comparisons between truck, rail, and pipeline are not distorted by differences in underlying network structure or data availability.

Transportation cost estimates for ammonia are sourced from [13], based on 2019 price levels across several onshore transport modes. Since the model requires a single distance-based cost rate to be multiplied by the Haversine distance, the mean cost across all available transport modes is computed and adopted as the baseline value. This aggregation ensures that the cost parameter is representative of the general onshore transport cost rather than being tied to a single transportation mode. The resulting cost trend is illustrated in Figure 5.2 and the full dataset underlying this calculation is provided in Appendix B.

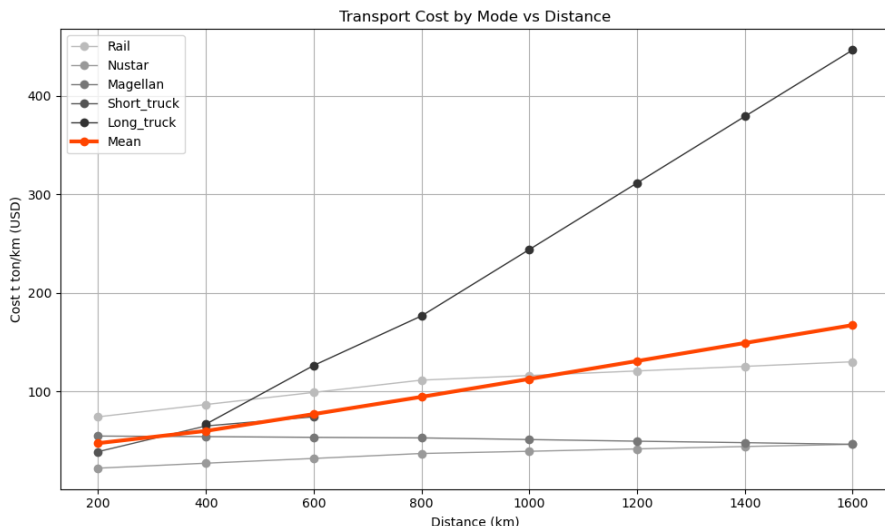


Figure 5.2: Onshore ammonia transportation cost by mode, 2019 [inserttransportref].

Two assumptions about onshore transport connections are made: a landmass restriction and a maximum distance threshold.

An initial approach considered restricting onshore transport to connections within the same region. However, this proved unrealistic in several cases. In Europe, cross-regional transport is both geographically feasible and commercially common, and a restriction on transportation across regions would limit connectivity in the model. Similarly, Africa is subdivided into East Africa, West Africa, and the Red Sea region in this model, and limiting transport strictly within these subregions produced unrealistic transportation outcomes.

To address this, a maximum onshore transport distance of 3,219 km (2,000 miles) is introduced, beyond which onshore transport is considered infeasible. This threshold is grounded in infrastructure reality: some of the world’s longest ammonia pipelines extend up to 2,000 miles [max_pipe], and since the onshore transport cost in this model represents an aggregated mix of transportation modes, including truck, rail, and pipeline, this is used as the upper bound. The distance is applied as a radius around each node, such that onshore transport is only permitted to nodes falling within this radius, regardless of node type. If there is more than this distance between two nodes C_{ij} is set to ∞ .

In addition to the maximum distance, onshore transport is restricted to nodes located on the same landmass. If nodes i and j reside on different landmasses, C_{ij} is set to ∞ . This restriction naturally handles island nodes: if a production or off take centre is located on an island, all flow to and from that node must be routed through a nearby transit node, hence via maritime shipping.

Several geographic corrections are applied where landmass connectivity does not reflect practical transport feasibility. Europe and Africa are technically connected via the Sinai Peninsula and the bridges crossing the Suez Canal; however, they are treated as separate landmasses in this model. This assumption reflects the limited accessibility of overland crossings in the Red Sea region, making this corridor an unrealistic route for ammonia transport at the scale considered here.

6 Initial model and results

6.1 2030 baseline scenario

As described in the previous sections, the model relies on several input data sources, each covering a different part of the network. Based on these, a base scenario is constructed for the year 2030. This horizon is chosen because green ammonia is not yet commercially viable at scale today, but is widely projected to become a meaningful part of the energy and industrial landscape by 2030 [14].

The production level in the base scenario is anchored to the IEA's 2030 forecast of global low-emissions ammonia output, which projects approximately 8 Mt of production capacity coming online by 2030 [1]. This global figure is distributed across the production locations described in Section 5.3. The resulting per node capacities define the upper bound on supply available to the model.

The ammonia demand for each industry is converted from production volumes and alternative fuel consumption using the factors described in Section 5.2 .

Applying the urea-to-ammonia conversion factor of 0.58 to the total global fertiliser production volume of 189 Mt yields the theoretical ammonia demand from the fertiliser sector:

$$189 \text{ Mt} \times 0.58 \approx 109.6 \text{ Mt NH}_3$$

This number of approximately 109.6 Mt represents the green ammonia demand that would arise if the entire global fertiliser industry were supplied with green ammonia.

Applying the ammonia intensity factor of 0.31 to the total global steel production volume of 1,886.9 Mt yields the theoretical ammonia demand from the steel sector:

$$1,886.929 \text{ Mt} \times 0.31 \approx 584.9 \text{ Mt NH}_3$$

This number of approximately 584.9 Mt represents the ammonia demand from the steel sector under a full transition to hydrogen-based DRI.

Applying the energy-equivalent substitution factor of 2.2 kg NH₃ per kg of fuel oil to the 2026 global bunkering volume of 249.65 Mt yields the theoretical ammonia requirement for marine shipping:

$$2.2 \times 249.65 \text{ Mt} \approx 549.23 \text{ Mt NH}_3$$

This corresponds to the ammonia volume required to replace global marine fuel oil consumption on an energy-equivalent basis.

The numbers above represent theoretical ammonia demand at each sector's current production level. To align the demand side with the 2030 production horizon, the projected annual growth rate of each industry is applied to scale its baseline ammonia demand forward to 2030. The growth rates used and the resulting 2030 demand volumes are summarised in Table 6.1.

Industry	Metric	Baseline	2030	Annual Growth	Source
Fertiliser	Ammonia demand [Mt]	109.6	119.3	1.4%	[15]
Steel	Ammonia demand [Mt]	584.9	617.3	0.95%	[16]
Shipping	Ammonia demand [Mt]	549.23	638.26	2.54%	[17]

Table 6.1: Projected annual growth in industrial production between the baseline year and 2030

This constitutes the supply and demand side of the 2030 base scenario. The following section presents the initial results obtained when the model is solved under these conditions.

6.2 Results base scenario

The base case is solved for 2030 with a total production capacity of 8 Mt, against a modelled global demand of 920 Mt. Production utilization reaches 100%, confirming the model operates in a deeply supply-constrained regime where every unit of available capacity is deployed. Only 0.87% of total demand is served, leaving 912 Mt unmet.

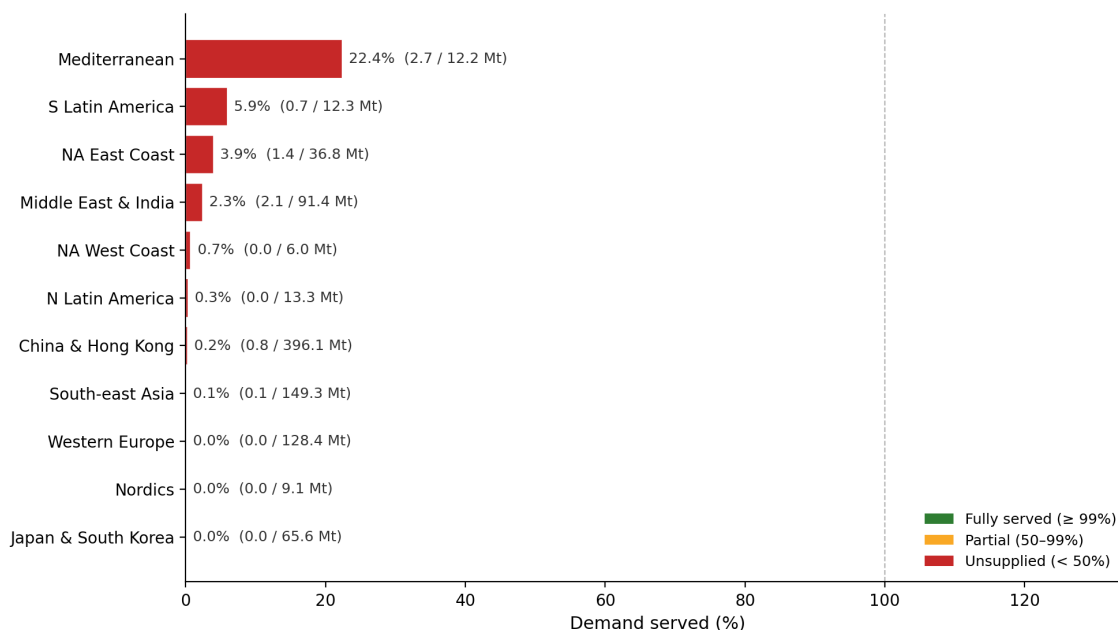


Figure 6.1: Regional coverage at 8 Mt supply

No single region is fully served, with the highest regional coverage reaching 22% in the Mediterranean. Most regions receive less than 5% of their demand, and several large demand centres including Western Europe (128 Mt), Japan & South Korea (66 Mt), and China & Hong Kong (396 Mt) receive effectively nothing. The relatively high Mediterranean share is explained by network geography: production nodes in Algeria, Mauritania, and Sudan route through the Gibraltar transit node, and a Mediterranean off take centre is near this hub, meaning it can be served at minimal additional transport cost. This is illustrated in Figure 6.2, which shows the base case flow network concentrated in the Africa-Mediterranean corridor.

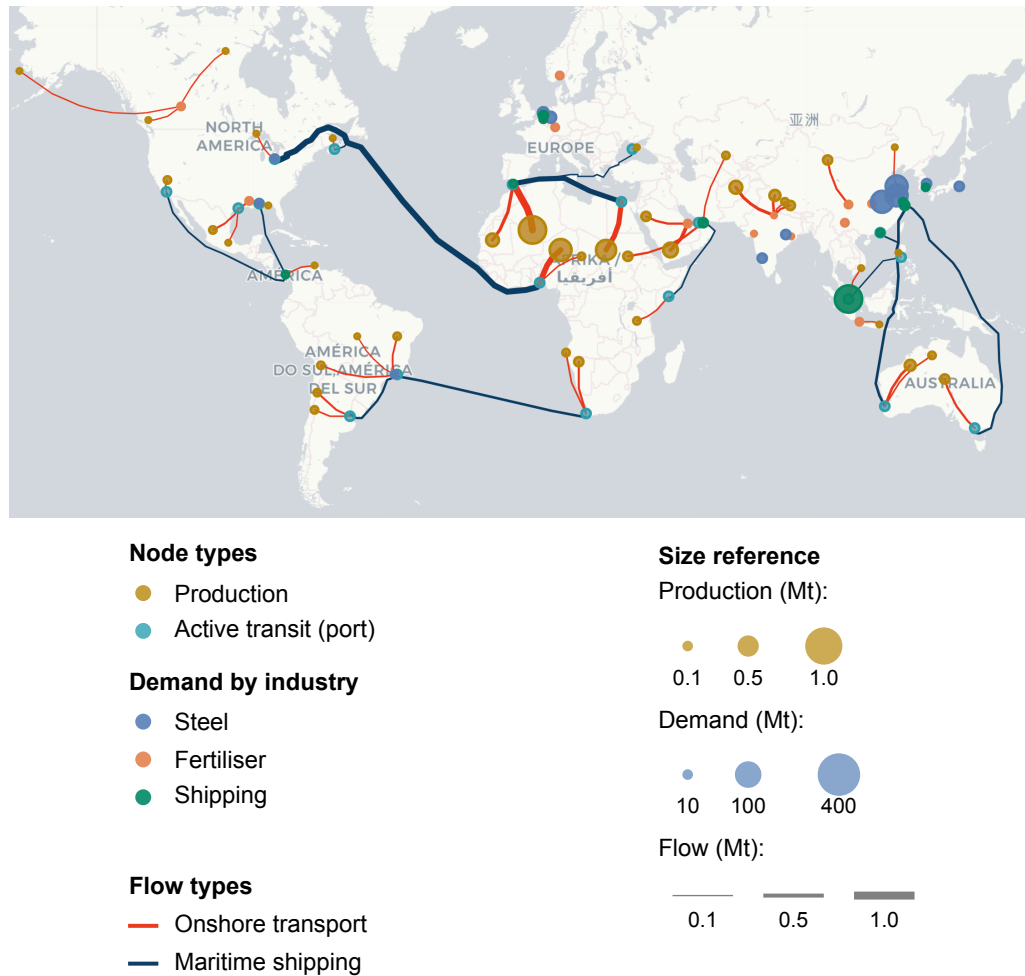


Figure 6.2: Base case flow network (8 Mt capacity, 2030).

6.3 Model refinement

The base case results reveal an unrealistic allocation of scarce supply to the shipping sector, which receives 47% of total delivered ammonia. In the absence of a global CO₂ pricing mechanism for maritime transport, shipping operators face an effective fuel cost well below the minimum green ammonia production cost of \$1,050/t modelled here.

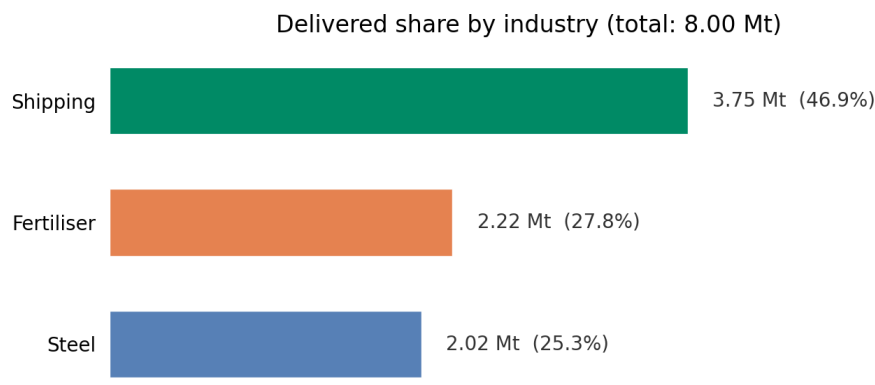


Figure 6.3: Enter Caption

While current geopolitical tensions, including the ongoing conflict affecting the Strait of Hormuz, have introduced significant volatility into HFO markets, green ammonia remains uncompetitive as a direct fuel substitute at current production costs without carbon pricing. (ved ikke om dette skal stå her?)

The optimizer allocates supply to shipping purely because the unmet demand penalty treats all sectors equally, not because shipping would realistically procure green ammonia at prevailing market prices.

Additionally, the initial model treats shipping demand as fixed at predetermined bunkering locations. This is an oversimplification: unlike industrial off take facilities, bunkering infrastructure is not geographically fixed and can follow the economics of where ammonia is cheapest to supply. We therefore make shipping demand variable, allowing the optimizer to determine both the volume and location of bunkering, subject to the revised penalty. This better reflects the flexibility of maritime refuelling and avoids artificially constraining the network around fixed port locations.

6.4 Capacity sensitivity and model validation

To verify that the model correctly transitions from supply-constrained to demand-satisfying behaviour, we solve an oversupply scenario with 1100 Mt of total production capacity - sufficient to exceed the 920 Mt of modelled demand. Together with the base case (8 Mt), these two scenarios bracket the feasible operating range of the model and allow us to confirm that key structural properties hold across regimes.

Table 6.2 summarises the results across both capacity levels.

Table 6.2: Capacity sweep summary

Capacity (Mt)	Produced (Mt)	Utilisation (%)	Demand (Mt)	Delivered (Mt)	Unmet (Mt)
8	8.00	100.0	920.5	8.00	912.5
1 100	913.3	83.1	920.5	920.5	0.0

At 8 Mt, production utilisation reaches 100% and only 0.87% of total demand is served, confirming that the model operates in a deeply supply-constrained regime. At 1100 Mt, all demand is fully satisfied and utilisation falls to 83%, indicating that the model correctly transitions to a demand-constrained regime with surplus capacity.

All regions are served at 100% coverage, confirming that the transport network is capable of connecting all production regions to all demand centres when capacity is not a binding constraint. No demand node remains unserved, verifying that the residual unmet demand observed in supply-constrained scenarios is attributable to capacity scarcity rather than network unreachability.

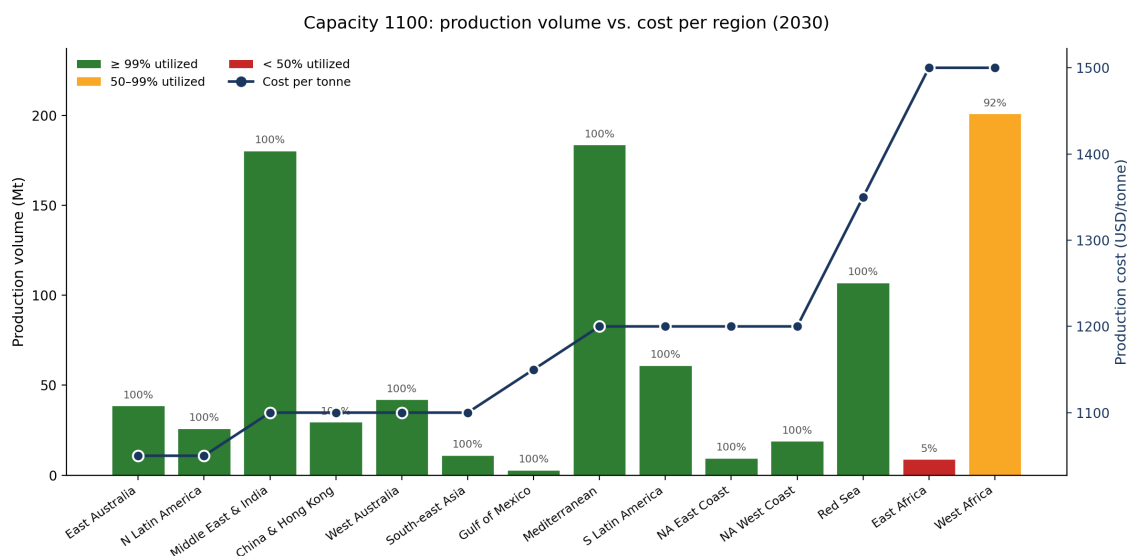


Figure 6.4: Production volume and cost by region at 1100 Mt capacity (2030)

The idle capacity is concentrated in the highest-cost production regions. East Africa (\$1,500/t) operates at only 5% utilisation, while West Africa (\$1,500/t) reaches 92%. All regions with production costs at or below \$1,350/t are fully utilised. This confirms that the optimizer follows cost-minimising dispatch: it exhausts cheaper production before drawing on expensive capacity, and leaves the most costly producers partially or fully idle once demand is satisfied. In contrast, the base case shows 100% utilisation across all regions regardless of cost, as the severity of the supply constraint means even the most expensive producers are preferable to paying the unmet demand penalty.

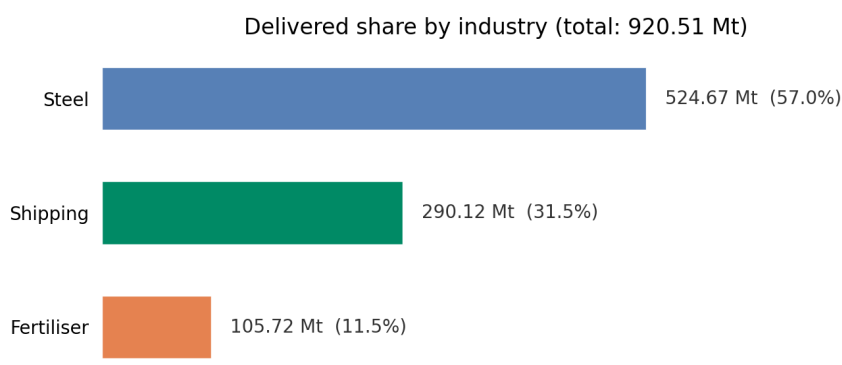


Figure 6.5: Delivered ammonia by sector at 1100 Mt capacity (2030)

Under full supply, the industry allocation reflects the underlying demand structure: steel accounts for 57% (525 Mt), shipping for 32% (290 Mt), and fertiliser for 12% (106 Mt). This contrasts with the base case, where shipping received 47% of the 8 Mt supply, a disproportionate share that motivated the model refinements discussed in Section ??.

These results validate two key model properties. First, the penalty parameter is correctly calibrated: all demand is served before any capacity is left idle, confirming that the penalty exceeds the cost of the most expensive viable production-to-off-take route. Second, the cost-minimising objective produces a physically intuitive dispatch order, cheap regions

are exhausted first, expensive regions are drawn upon last, consistent with the expected behaviour of a minimum-cost network flow formulation.

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A Calculations

A.1 Ammonia demand

A.1.1 Steel industry

Hydrogen-based steel production requires approximately 0.055 tons of hydrogen per ton of steel produced. [10] The hydrogen mass fraction in ammonia is [11]:

$$\frac{3}{17}$$

This corresponds to:

$$1 \text{ ton NH}_3 = 0.176 \text{ ton H}_2$$

The ammonia requirement is then

$$\frac{0.055}{0.176} \approx 0.31$$

Therefore, approximately a factor of 0.31 is used to convert the ammonia used in steel.

B Transportation cost for ammonia

Distance (km)	Rail	Nustar Pipeline	Magellan Pipeline	Short-dist. truck	Long-dist. truck	Mean
200	74.40	22.32	54.87	39.06	–	47.6625
400	86.86	27.34	54.31	65.10	66.96	60.1152
600	99.14	32.18	53.57	74.40	126.48	77.1528
800	111.60	37.20	53.01	–	176.70	94.6275
1000	116.25	39.43	51.34	–	244.03	112.7625
1200	120.90	41.85	49.66	–	311.55	130.9905
1400	125.55	44.27	48.17	–	379.07	149.2650
1600	130.20	46.50	46.50	–	446.40	167.4000

Table B.1: Transport Costs by Mode and Distance (multiplied by 18.6)

Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. This text should contain all letters of the alphabet and it should be written in of the original language. There is no need for special content, but the length of words should match the language.

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